High school · number theory

Congruences and last digits

Worked solutions

Exercise 1

Find the remainder of -37 modulo 8.

$-37 = 8 \times (-5) + 3$, so the remainder is 3.

Exercise 2

Find the last digit of 7^{2026}.

The cycle is 7,9,3,1 and $2026 \equiv 2$ (mod 4). The last digit is 9.

Exercise 3

Prove that an integer square is never congruent to 2 modulo 3.

An integer is congruent to 0,1 or 2. The corresponding squares are 0,1 and $4 \equiv 1$ modulo 3.

Exercise 4

Solve $3x \equiv 1$ (mod 7).

$3 \times 5 = 15 \equiv 1$. Multiplying by 5 gives $x \equiv 5$ (mod 7). Conversely, every such value works.

Problem: complete reasoning

An integer N is congruent to 2 modulo 3 and to 3 modulo 5. Write $N = 3k+2$, determine k modulo 5, then give all solutions modulo 15. Which solutions from 0 to 100 inclusive are even?

$3k + 2 \equiv 3$ (mod 5), so $3k \equiv 1$ and $k \equiv 2$ (mod 5), since $3 \times 2 \equiv 1$. Thus $N = 15t + 8$. In [0,100] the values are 8,23,38,53,68,83,98. The even values are 8,38,68,98. Substitution verifies both congruences for every integer t.